Alexander Zhang

437-362-4903 | alexanderz.zhang@mail.utoronto.ca | linkedin.com/in/azutsc | github.com/CodingRookieA

Education

University of Toronto Scarborough Scarborough, ON
*Honours B.Sc. Specialist in Computer Science Co-op – **GPA: 3.89/4.0*** *Sep. 2023 – Present*

- Awards: Renewable Entrance Scholarship ($3,000, 2023–2025); Dean's List (2023–2025); Top 25% — Euclid Mathematics Contest

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, Bash, SQL, HTML/CSS
Frameworks & Libraries: React, Node.js, Express, Flask, Socket.io, WebRTC, PyTorch, Hugging Face Transformers, scikit-learn, Jest, JUnit, pandas, NumPy
Databases: MongoDB, MongoDB Atlas (Vector Search), PostgreSQL, Firebase, NeDB
DevOps & Tools: Docker, Nginx, Jenkins, Git/GitHub, Linux CLI, GitHub Actions, Azure DevOps, Jira
AI/ML: RAG pipelines, LLM prompt engineering, Gemini API, OpenAI API, DistilBERT, n8n, Power Automate

Experience

Research and Strategy Assistant (Co-op) Jan. 2026 – Apr. 2026
Uphouse Inc. *Winnipeg, MB*

- Built automated workflows using Power Automate and n8n with conditional logic, input normalization, and null-handling to eliminate daily manual deadline tracking for the team.
- Integrated Trello with n8n via webhooks to trigger client-facing feedback forms on card completion, applying event filtering to extract relevant fields from raw webhook payloads.
- Analyzed AI visibility gaps across four industry verticals using Profound, driving visibility improvement from near-zero to approximately 50% within one month.

IT Analyst (Co-op) June 2025 – Aug. 2025
Employment and Social Development Canada (ESDC) *Toronto, ON*

- Developed and automated internal workflows using Microsoft Power Platform (Power Automate, Power Apps, MS Forms), reducing manual processing time by 50%.
- Conducted structured testing on internal OCR-based AI tools using a custom evaluation framework in Excel, contributing directly to tool selection decisions for document automation.
- Managed Agile sprints via Azure DevOps, creating and maintaining work tasks and standardized test suites for client-facing demonstrations.

Projects

Finwise — AI Financial Advisor | *React, Node.js, MongoDB Atlas, Python, Gemini API, Jest* Jan. 2026 – Present

- Built a full-stack AI mutual fund and ETF advisor (MERN stack) with profile-aware prompt orchestration, live market data integration, and session-based multi-turn conversation history.
- Designed a full RAG pipeline: cleaned, chunked, and embedded investment articles into MongoDB Atlas with 3072-dim vector search indexing to ground AI responses and reduce hallucinations.
- Built a multi-signal intent classifier that routes queries to article retrieval, fund data, or ETF data — detecting continuation turns, enforcing scope gating, and controlling token cost.
- Engineered a nightly web-scraping pipeline across major Canadian fund families (TD, RBC, Mawer, Fidelity) to extract NAV, MER, and risk ratings into a structured MongoDB collection.
- Architected the backend with separation of concerns using factory-based dependency injection; wrote Jest unit tests mocking DB and AI clients to validate session ownership and response contracts.
- Implemented CI/CD via GitHub Actions with lint, test, and security gates; built and pushed Docker images via Buildx with environment-specific secrets injection.

MindBit — AI Social App | *React, Node.js, TypeScript, Socket.io, WebRTC, Docker, MongoDB* Sep. 2025 – Dec. 2025

- Built a full-stack social app with MBTI personality assessment and AI game recommendations using React, Node.js/TypeScript, and the OpenAI and Perplexity APIs.

- Implemented real-time chat (Socket.io), peer-to-peer video chat (WebRTC with Socket.io signaling), salted/hashed local auth, Steam OAuth, and TOTP-based 2FA — all with persistent cross-session history.
- Containerized with Docker behind an Nginx reverse proxy with automated Let's Encrypt HTTPS; deployed a distributed MongoDB setup splitting sensitive credentials from storage-heavy data across Atlas and a VM.
- Coordinated backend/frontend integration with a partner via Git branching and PR reviews, reducing merge conflicts by 90%+; used GitHub Copilot to accelerate development and debugging by 40%+.
- Added web scraping to verify AI-recommended games have correct Steam links, images, and descriptions, ensuring recommendation accuracy before surfacing results to users.

ML Sentiment Analysis | *Python, PyTorch, Hugging Face Transformers, scikit-learn* Dec. 2025

- Built an end-to-end NLP pipeline fine-tuning a DistilBERT classifier for movie review sentiment, handling data preprocessing, tokenization, training, and evaluation within a unified PyTorch/Hugging Face workflow.
- Benchmarked the Transformer model against a TF-IDF + Logistic Regression baseline using scikit-learn metrics, quantifying the performance advantages of the deep learning approach.
- Developed a reproducible inference workflow to load the trained model and generate sentiment predictions on unseen text, enabling practical deployment outside the training environment.

CI Pipeline with Jenkins & GitHub | *Jenkins, Docker, Python, Pytest, Flake8, GitHub* Dec. 2025

- Built a fully automated CI pipeline using Jenkins and Docker: triggered branch-level builds via GitHub webhooks, enforced Flake8 linting and Pytest suites in isolated virtual environments, and blocked merges on failures via GitHub branch protection rules.
- Resolved CI execution failures related to workspace permissions, Python module resolution, and Jenkins security constraints, restoring reliable automated validation across all branches.

Web Gallery Application | *Node.js, JavaScript, HTML/CSS, NeDB, Docker, Nginx* Sep. 2025 – Dec. 2025

- Built a responsive image-sharing app with a Node.js/NeDB backend exposing RESTful CRUD endpoints, session-based auth/authorization, and mobile-first frontend.
- Refactored into a decoupled client–server architecture and deployed with Docker behind an Nginx reverse proxy to secure traffic and handle CORS.

DBMS Investing Application | *PostgreSQL, SQL, Google Cloud, AWS EC2* Sep. 2025 – Dec. 2025

- Designed a database-backed investing app with a BCNF-normalized PostgreSQL schema (E/R diagram → relation sets → BCNF), deployed across Google Cloud VM and AWS EC2 for consistent availability.
- Developed and implemented the database using PostgreSQL with SQL for efficient data management and querying, improving query performance and maintainability.

Linux and Systems Programming | *C, Bash, Linux CLI* Jan. 2025 – Apr. 2025

- Built a C-based system monitoring tool displaying active processes, CPU, and memory usage in real time using multi-threading and multi-processing.
- Automated file management and process monitoring via Bash scripts; wrote technical documentation covering usage, expected inputs/outputs, and common errors.

Planetze — Android Eco-Tracking App | *Java, Android Studio, Firebase, JUnit, Jira* Sep. 2023 – Dec. 2023

- Led a team of 5 as Scrum Master and Project Manager, managing workflow in Jira, writing user stories, and coordinating parallel feature branches across the full development lifecycle.
- Implemented the app's backend logic in Java, including user data processing and Firebase authentication and storage integration for secure credential management.
- Refactored the login system to follow the MVP design pattern and validated all functionality with JUnit test cases, improving maintainability and achieving full test coverage.

FriendFind | *HTML, JavaScript, MongoDB Atlas* | *Hack The Valley 9* Oct. 2024

- Built a relationship-building web app in 48 hours with a team of 3, enabling users to create profiles and post/respond to questions to encourage connection.
- Presented a live demo to a three-judge panel; received positive feedback including interest in pitching the prototype through UTSC's SOCIETAL entrepreneurship program.